

MANGALORE REFINERY AND PETROCHEMICALS LIMITED

Inspection Department - Equipment Condition Report

Report No.	INSP-2026-0631
Equipment Tag	E-204B (Crude Preheat Exchanger, shell side)
Unit	Crude Distillation Unit - 2
Line	8in-SS-2207
Inspection Date	09 September 2026
Technique	Ultrasonic thickness survey, visual, dye penetrant
Inspector	S. Prabhu, Level II UT / PT

FINDINGS

1. Shell wall thickness measured at 11.4 mm against a design minimum of 12.5 mm at grid reference C-7. BELOW RETIREMENT LIMIT.
2. Channel head gasket shows external weepage at the 4 o'clock position. Estimated leak rate 6 drops/minute.
3. Tube bundle fouling factor estimated at 0.0009 m2K/W against a design allowance of 0.0004 m2K/W. Duty shortfall observed.
4. Temperature indicator TI-412 reading 219 deg C on the shell outlet, exceeding the 205 deg C high alarm setpoint.
5. Pressure gauge PG-118 isolation valve found seized. Gauge cannot be isolated for calibration.
6. Insulation cladding damaged over approximately 1.2 m of the shell, with visible corrosion under insulation at two locations.

RECOMMENDATION

Remove E-204B from service and schedule for shutdown. Shell plate repair required at grid C-7. Renew channel head gasket. Bundle to be withdrawn and hydro-jetted. Replace PG-118 isolation valve and recalibrate TI-412 against a certified reference.

Criticality: HIGH

Next Inspection: 09 December 2026

ULTRASONIC THICKNESS SURVEY - E-204B SHELL

Report INSP-2026-0631 | Nominal 14.0 mm | Design minimum 12.5 mm

Grid	Measured	Loss	Rate	Status
A-1	13.6 mm	0.4 mm	0.04 mm/yr	ACCEPTABLE
A-4	13.2 mm	0.8 mm	0.08 mm/yr	ACCEPTABLE
B-2	12.9 mm	1.1 mm	0.11 mm/yr	MONITOR
B-7	12.7 mm	1.3 mm	0.13 mm/yr	MONITOR
C-3	12.6 mm	1.4 mm	0.14 mm/yr	MONITOR
C-7	11.4 mm	2.6 mm	0.26 mm/yr	BELOW LIMIT
D-1	13.1 mm	0.9 mm	0.09 mm/yr	ACCEPTABLE
D-5	12.8 mm	1.2 mm	0.12 mm/yr	MONITOR

NOTES

Survey performed on 10 years in service. Corrosion rate at C-7 is the governing value. Remaining life at C-7 is negative against the design minimum; the location requires immediate assessment.

All readings taken with a calibrated UT gauge, couplant applied.
Surface prepared by wire brushing prior to measurement.